

CO₂ and Photosynthesis

Pupil sheet

Name: _____

Date: _____

1. What I see on the screen

The screen shows four cards: **CO₂** (ppm), **Light** (lx), **Temperature** (°C) and **Humidity** (%). The temperature is not decorative: **if the lamp heats the jar we have changed two factors at once** and the series is no longer clean.

What I read is not the value but the gradient. How much air was in the jar at closing does not matter. What matters is *how much the CO₂ changes in a minute*, measured over at least **five minutes**.

The sensor does not see photosynthesis. It sees the *difference* between photosynthesis and respiration — the net rate. A flat line does not mean “nothing is happening”.

2. What I do on my own

First I predict. In bright light the CO₂ will: ☐ ↑ ☐ ↓ ☐ → In darkness it will: ☐ ↑ ☐ ↓ ☐ →

Is there a light level at which the line stays flat? ☐ yes ☐ no

because _____

The rig: plant _____ jar _____ L lamp at _____ cm

The two basic experiments. Worked example: 820 → 720 ppm in 10 min ⇒ $(720 - 820)/10 = -10$ ppm/min.

Condition	CO ₂ start	CO ₂ end	Minutes	Gradient (ppm/min)
Control, empty jar				
Bright light				
Complete darkness				

The intensity series. I check the temperature at every step.

Shading	Light (lx)	Gradient (ppm/min)	T (°C)
No sheet			
One sheet			
Two sheets			
Three sheets			
Darkness	0		

The graph. I place the points. **The compensation point is where the curve crosses the thickened zero line.**

Compensation point $\approx$ _____ lx. Check: the value at 0 lx should be the respiration rate.

3. The conclusion I write

I predicted that, and the data show that _____

When the line was flat, what was happening in the leaves was _____

I cannot conclude anything about the planet's forests because _____

Exit ticket

What is happening in the leaves when the CO₂ line is flat?
